Validation of Two-Box Spacer Modeling Across Spacer Widths for Fenestration Thermal Analysis

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Abstract

Accurate thermal modeling of window spacers is difficult to parameterize, validate, and is computationally intensive due to complex geometries requiring fine mesh resolution. Detailed spacer geometries require local element sizes on the order of 0.05-0.2 mm within high-conductivity regions to accurately represent spacer models. Two-box spacer modeling can simplify comparison and validation of spacer models, while reducing the computation requirements in full-window simulations. This paper evaluates two simplified two-box spacer modeling methodologies; one in which a new model is generated for each spacer width and another where a two-box model is calibrated at a nominal width and then resized across each spacer width. The study evaluates 3 representative spacer families across 25 discrete widths (8-20 mm) comprising 900 full-frame simulations. Results demonstrate that full-window U-factor deviations remain within approximately 0.008 $W/m^2 \cdot K$ for spacer widths between 8 mm and 20 mm when two-box models are calibrated at each width, supporting the validity of the method for engineering and regulatory applications. The findings suggest a pathway toward standardized, defensible spacer abstraction suitable for both North American and European modeling contexts.

Introduction

The thermal performance of insulated glass units (IGUs) is strongly influenced by spacer systems, particularly through their contribution to the edge-of-glass heat transfer. Attempts to simplify detailed spacer models have been made, with significant adoption in European fenestration modeling methodologies, such as ISO 10077-2. These approaches have been adopted in North America, though to a lesser extent, in methodologies such as Passive House.

Simplified spacer modeling methods offer several advantages, including reduced solver mesh complexity, improved consistency when comparing models, and support for parametric analysis methods such as the NFRC Commercial Trendline Approach (CTA). Two-box modeling is well understood and generally accepted as equivalent to detailed modeling for U-factor ratings. However, the impacts of spacer width on thermal performance and condensation failure modes have not been studied in depth. This paper seeks to clarify these impacts and evaluate whether nominal-width calibrated two-box spacer models can replace their detailed model counterparts while validating their results for U-factor, CI edge, and CI frame.

This paper seeks first to validate two-box modeling across varying spacer widths for both U-factor and Condensation Index. It then examines whether the two-box model can be further simplified by calibrating the effective thermal conductivity at a nominal spacer width and representing all spacer widths with the same two-box model. All simulation results for U-factor and Condensation Index were found using LBNL THERM and the modeling methodologies from NFRC (100, 200, 500).

Background and Related Work

Previous investigations, including work conducted at Lawrence Berkeley National Laboratory, have explored simplified spacer representations constrained to specific geometric assumptions, materials, and nominal spacer widths. In some cases, model construction was limited to a fixed spacer width (e.g., 6mm) and a single sealant material, with the implicit assumption that the abstraction captures dominant thermal behaviours independent of spacer-specific details.

Such constraints raise important questions regarding generality. Spacer systems vary widely in width, material composition, and internal geometry. A modeling approach that remains valid across these variations would provide substantial practical value. The present study builds on this foundation by explicitly testing width sensitivity while retaining a consistent two-box abstraction.

Methodology

Detailed Spacer Models

Detailed spacer models were constructed for multiple spacer families, spanning low, medium, and high effective thermal conductivities. These models served as the reference baseline for comparison and were simulated both independently and within a full window context to capture component level and system level impacts.

Two-Box Spacer Models

Two-box spacer models were generated as simplified geometric representations intended to replicate the overall thermal behaviours of the corresponding detailed spacer. In the first study, two-box models were created, and calibrated for each spacer width. In the second study, a two-box model was created and calibrated at a nominal width of 8 mm and then subsequently resized to represent all other spacer widths without further recalibration.

Both the detailed and two-box models were evaluated using identical boundary conditions, glazing systems, frame geometries, and solver settings to ensure consistent comparison. Model generation and calibration were automated to ensure consistency and repeatability across the full parameter space. Automated model generation and calibration scripts included validation checks for boundary condition and material property assignments. The default THERM spacer boundary conditions were used to calibrate the models, with an exterior temperature of -18.0 C and interior temperature of 21.0 C. The default heat transfer coefficient of 9999.0 $W/m^2 \cdot K$ was used in both the interior and exterior to approximate isothermal boundary conditions and ensure any heat transfer through the spacer is governed solely by conduction. As a result of this, component-level

U-factor values and associated errors can become numerically large and should be interpreted as calibration metrics rather than physically representative system-scale heat transfer coefficients. These isothermal boundary conditions are used solely for calibration; all full-window results use NFRC 100 standard boundary conditions.

Calibration was performed by constructing a two-box model with representative geometry derived from the detailed spacer. The lower box corresponded to the continuous portion of the detailed spacer geometry. The upper box represented the remaining portion of the spacer geometry and was assigned an initial thermal conductivity of 0.1 W/m·K. We iteratively adjusted the upper box's thermal conductivity using the Bisection method until the two-box model's component-level U-factor matched the detailed model within 0.01 $W/m^2 \cdot K$. Bisection iterations were terminated when the absolute difference between the two-box and detailed component-level U-factor fell below 0.01 $W/m^2 \cdot K$. All cases converged within 17 iterations. The selected tolerance ensures calibration error remains well below full-window sensitivity.

Spacer Families and Test Matrix

Three spacer families were evaluated in this study:

- Composite single seal spacers
- Metal single seal spacers
- Metal double seal spacers

Spacer widths ranged from 8 mm to 20 mm in 0.5 mm increments, resulting in 25 discrete spacer widths per family. For each spacer width, both a fully detailed spacer model and a corresponding two-box model were evaluated.

All two-box calibrations met convergence criteria within a tolerance of 0.01 $W/m^2 \cdot K$. In total, the study comprised a large set of paired detailed and two-box simulations across spacer family and width variations.

Frame and Glazing System

All spacer models were evaluated within a wood frame with no air cavities to reduce error introduced by frame convection variations. All spacers were evaluated with double glazed, air filled insulated glazing units of varying gap width. Glazing makeups and frame materials were held constant across all simulations to isolate spacer effects.

Results

A total of 25 spacer widths were evaluated for each of three spacer families using both detailed and two-box modeling approaches. Each spacer configuration was inserted into its corresponding frame geometries, including the head, jamb, and sill models. This resulted in 450 full-frame simulations for the width calibrated study. An additional 450 simulations were completed for the nominal-width calibrated study, yielding a total of 900 full-frame simulations.

For each standalone spacer model, the U-factor and calibrated upper box thermal conductivity were recorded. For full-frame simulations, overall U-factor, edge Condensation Index, and frame Condensation Index were evaluated.

Two-box models calibrated at each width

When two-box models were independently calibrated for each spacer width, convergence to within 0.01 $W/m^2 \cdot K$ of the detailed U-factor was achieved within 17 iterations using the bisection method for all spacer families.

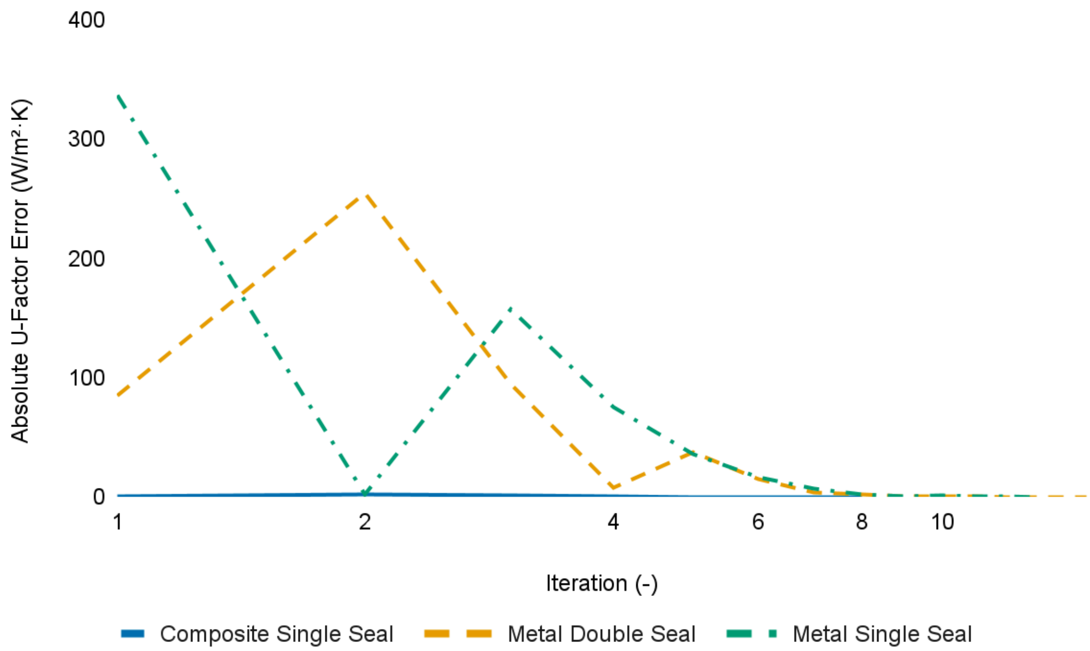

Figure 1. Absolute U-Factor Error As a Function of Iteration Number During Two-Box Spacer Calibration.

Absolute U-factor error as a function of iteration number during two-box spacer calibration using the bisection method. Results are shown for representative 16 mm spacers from three spacer families: composite single seal, metal double seal, and metal single seal. Error is defined as the absolute difference between the two-box model U-factor and the corresponding detailed spacer model. The y-axis is shown on a logarithmic scale to illustrate convergence behaviours across several orders of magnitude. All spacer families converge to within 0.01 W/m²·K within 17 iterations.

At the component level, differences in U-factor between detailed and two-box spacer models were negligible across the full width range. No systematic drift with spacer width was observed for any spacer family.

When inserted into the full-frame assemblies, width-calibrated two-box models showed strong agreement with detailed spacer models across all evaluated metrics. Full-window absolute U-factor, CI Edge, and CI Frame error remained consistent across spacer widths. The largest deviations were observed in absolute CI Frame error for the high conductivity spacer family; however, these deviations were stable with spacer width and did not exhibit cumulative growth.

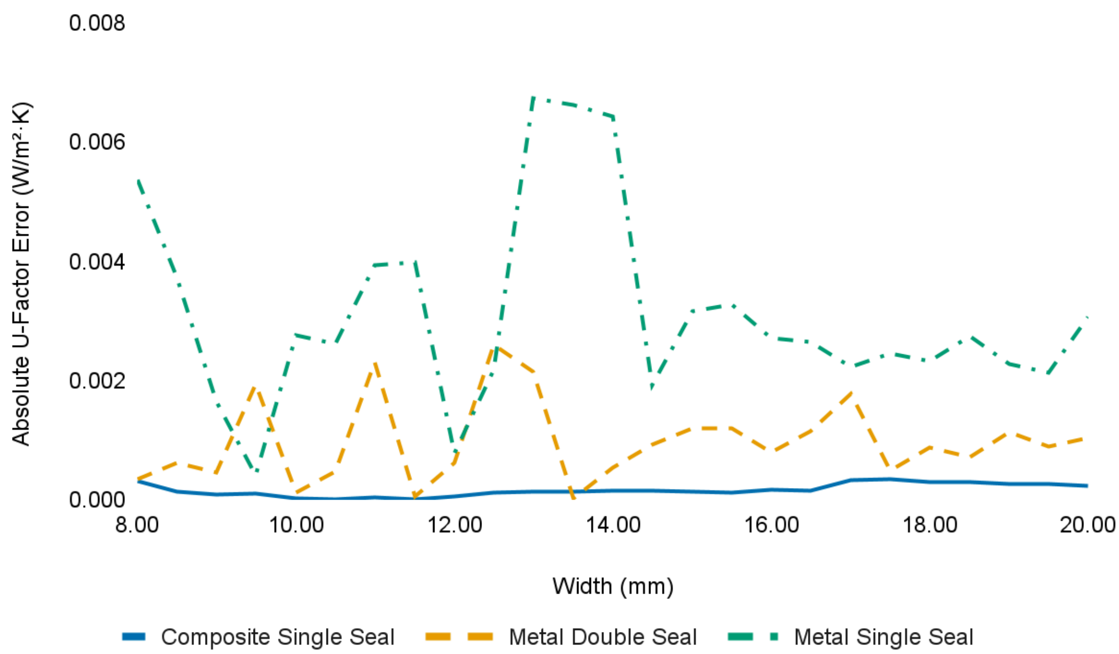

Figure 2: Full-Window Absolute U-Factor Error Across Spacer Widths

Absolute error in full-window U-factor between width-calibrated two-box spacer models and detailed spacer models for three spacer families (composite single seal, metal double seal, and metal single seal). Errors remain below 0.008 W/m²·K across the 25 tested spacer widths (8 - 20 mm), demonstrating that the two-box models accurately reproduce overall thermal performance when integrated into head, jamb, and sill frame assemblies.

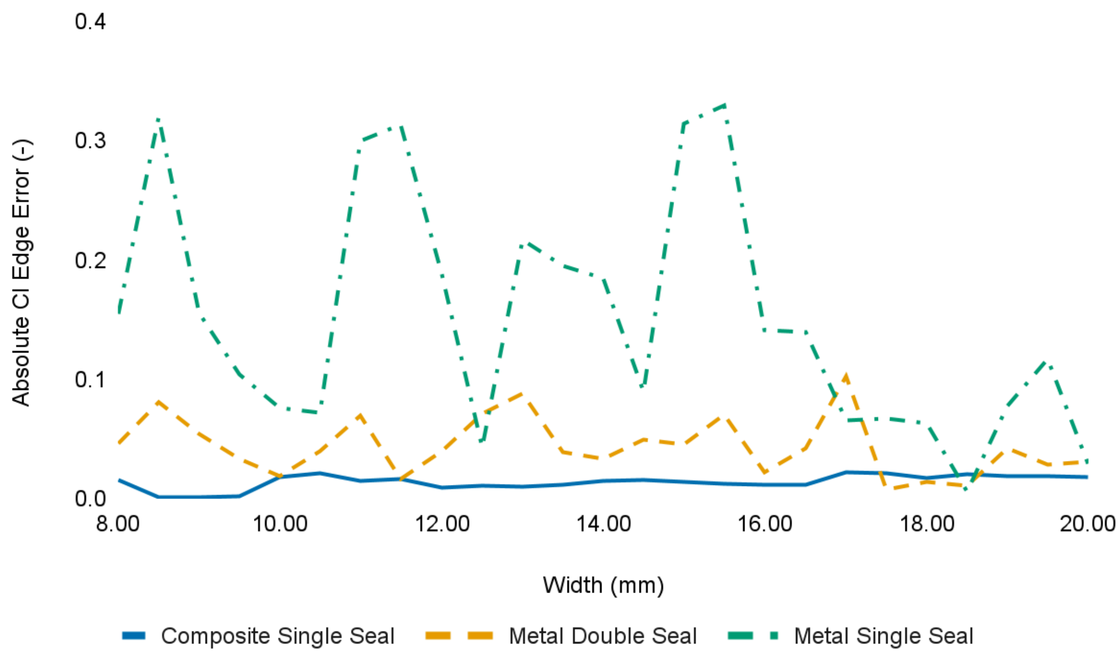

Figure 3: Full-Window Absolute CI Edge Error Across Spacer Widths

Absolute error in condensation index at the edge-of-glass (CI edge) between width-calibrated two-box spacer models and detailed spacer models for three spacer families. Errors remain below 0.4 (-) across all tested spacer widths, showing that two-box models maintain high fidelity for edge thermal behaviour within full window assemblies.

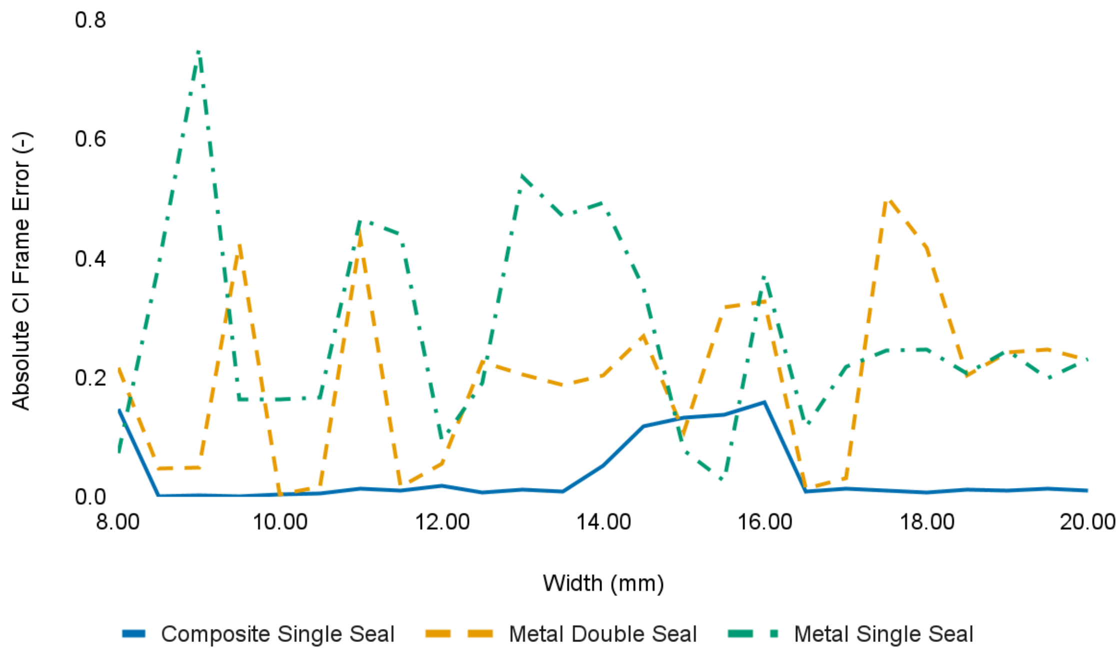

Figure 4: Full-Window Absolute CI Frame Error Across Spacer Widths

Absolute error in condensation index at the frame (CI frame) between width-calibrated two-box spacer models and detailed spacer models for three spacer families. Errors remain below 0.8 (-) across all tested spacer widths, showing that two-box models maintain high fidelity for frame thermal behaviour within full window assemblies.

Two-box models calibrated at a nominal width

In the second study, two-box models were calibrated at a nominal spacer width of 8 mm and subsequently resized to represent spacer widths ranging from 8 mm to 20 mm without further recalibration. The 8 mm nominal width was chosen because it represents the lower bound of the evaluated range. Calibrating at minimum width produces conservative (higher) U-factor predictions as width increases, consistent with regulatory preference of conservative estimates.

At the component level, two-box models calibrated at a nominal spacer width exhibited increasing U-factor error as spacer width deviated from the calibration point. For all spacer families, the absolute U-factor error increased rapidly at smaller width deviations, followed by a reduced rate of error growth at larger spacer widths. In some cases, the error plateaued or marginally decreased at the largest widths evaluated.

The magnitude and sensitivity of the error varied systematically by spacer family. Low-conductivity composite spacers showed relatively small absolute errors across the full width range, while metal spacer systems exhibited substantially higher sensitivity to width variation. The medium-conductivity spacer family demonstrated absolute U-factor errors exceeding values typically acceptable for regulatory modeling when applied outside the calibration width (> 0.01 W/m²·K). This behaviour indicates that two-box models calibrated at a single nominal width are not generally transferable across spacer widths without recalibration, particularly for high-conductivity spacer systems.

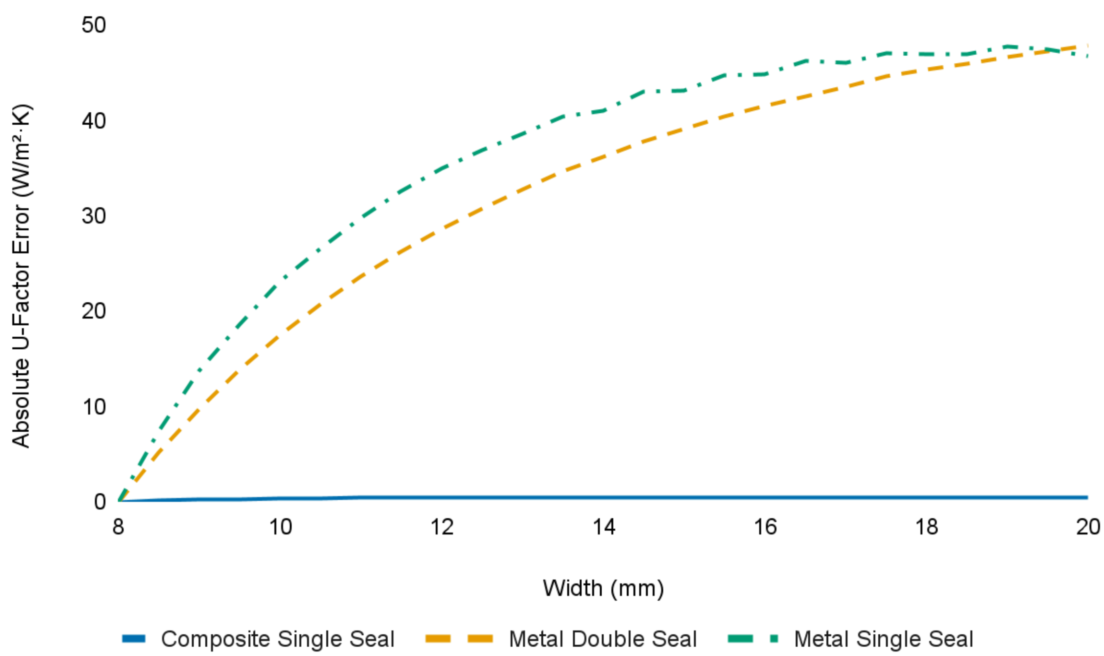

Figure 5: Absolute U-Factor Error for Nominal-Width-Calibrated Two-Box Spacer Models as a Function of Spacer Width.

Absolute U-factor error between detailed spacer models and two-box spacer models calibrated at a nominal width of 8 mm and subsequently resized without further recalibration. Results are shown for composite single seal, metal single seal, and metal double seal spacer families across spacer widths ranging from 8 mm to 20 mm. All spacer families exhibit increasing error as spacer width deviates from the calibration width, with rapid error growth at smaller deviations followed by reduced sensitivity at larger widths. The magnitude of the error and its sensitivity to width variation increase with spacer effective thermal conductivity.

When incorporated into full-frame simulations, nominal-width calibrated two-box models exhibited increased deviation relative to the width-calibrated case across all evaluated metrics. Full-window U-factor, CI edge, and CI frame errors increased with spacer width for all spacer families. The medium-conductivity spacer family exhibited the largest deviations over the evaluated width range, while the composite spacer family showed the lowest sensitivity.

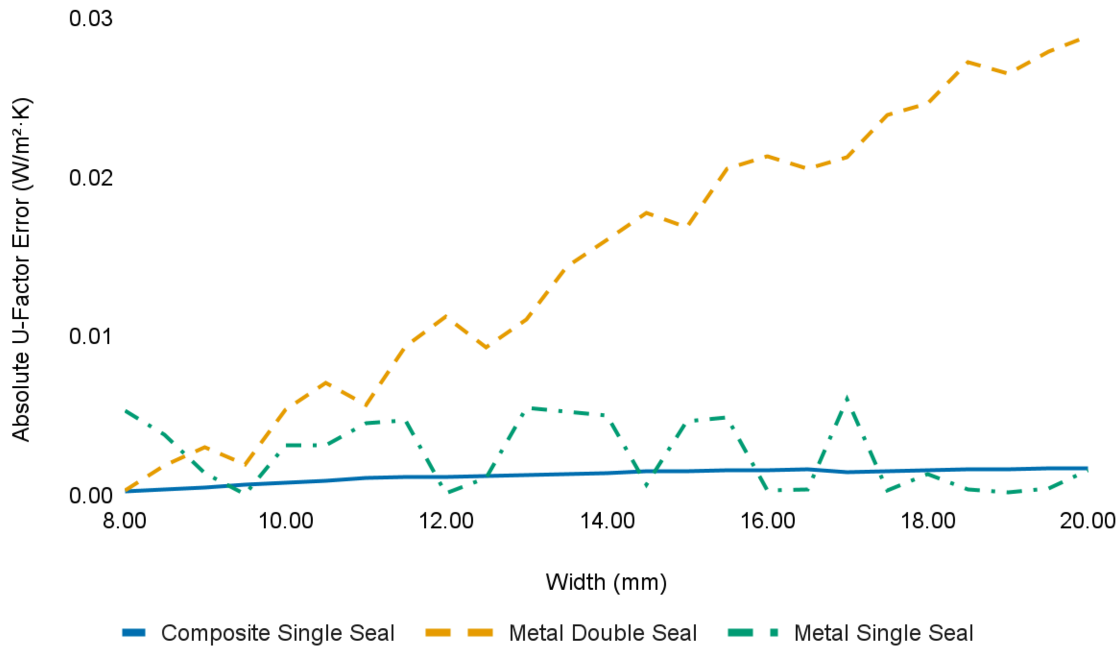

Figure 6: Full-Window Absolute U-Factor Error as a Function of Spacer Width for Nominally Calibrated Two-Box Models.

Full-window absolute U-factor error for nominal-width (8 mm) calibrated two-box spacer models evaluated across spacer widths from 8 mm to 20 mm. Errors are shown for three spacer families: composite single seal, metal double seal, and metal single seal.

The metal double seal spacer exhibits a steady increase in absolute full-window U-factor error with increasing spacer width, indicating sensitivity to width variation when applied outside the calibration point. Past 13 mm (+5 mm deviation), the metal double seal spacer exceeds the 0.01 $W/m^2 \cdot K$ tolerance, confirming that nominal-width calibration is unsuitable for this spacer family. In contrast, the composite single seal and metal single seal spacers maintain very low absolute errors across the full width range, with errors remaining near zero and showing no systematic growth with spacer width.

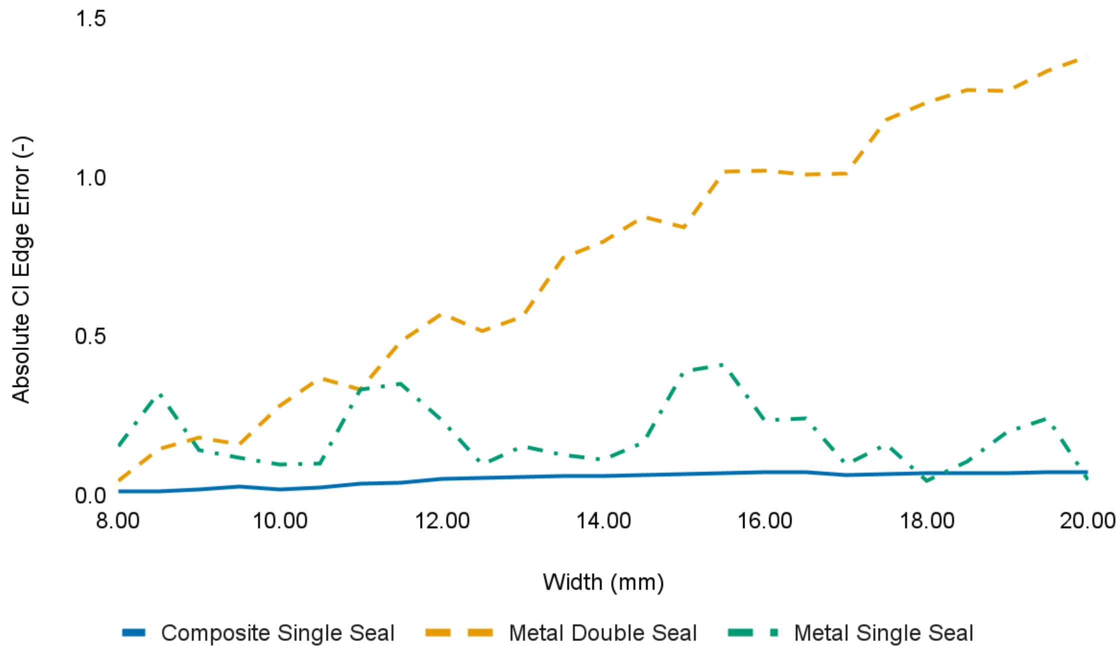

Figure 7: Full-Window Absolute CI Edge Error as a Function of Spacer Width for Nominally Calibrated Two-Box Models.

Full-window absolute CI edge error for nominal-width (8 mm) calibrated two-box spacer models evaluated across spacer widths from 8 mm to 20 mm. Errors are shown for composite single seal, metal double seal, and metal single seal spacer families.

The metal double seal spacer exhibits a strong and sustained increase in CI edge error with increasing spacer width, indicating high sensitivity to width variation when applied outside the calibration point. In contrast, the composite single seal spacer shows relatively small CI edge errors with only a modest increase across the evaluated range. The metal single seal spacer demonstrates non-monotonic behaviours, with elevated errors at select widths but no consistent growth trend over the full range.

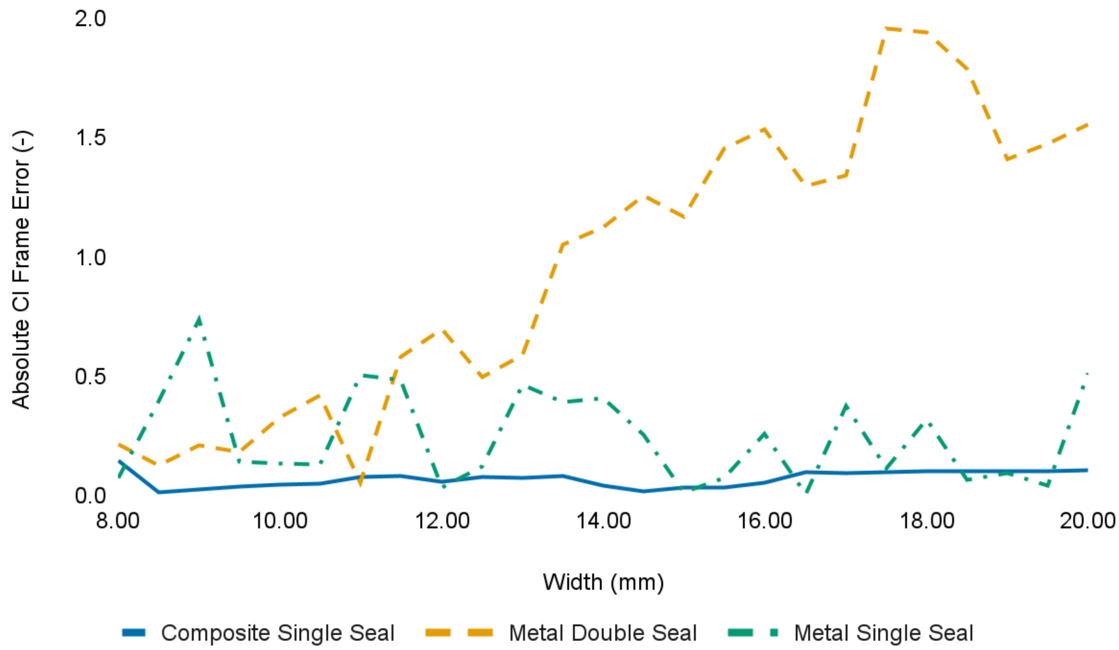

Figure 8: Full-Window Absolute CI Frame Error as a Function of Spacer Width for Nominally Calibrated Two-Box Models.

Full-window absolute CI frame error for nominal-width (8 mm) calibrated two-box spacer models evaluated across spacer widths from 8 mm to 20 mm. Results are shown for composite single seal, metal double seal, and metal single seal spacer families.

CI frame error exhibits pronounced spacer-family dependence and non-monotonic behaviour with spacer width. The metal double seal spacer consistently produces the largest CI frame errors across the evaluated width range, with error magnitude generally increasing at larger spacer widths. In contrast, the composite single seal spacer maintains relatively low CI frame errors with limited sensitivity to spacer width. The metal single seal spacer demonstrates highly variable behaviour, with localized peaks in error but no sustained trend with increasing width.

Discussion

This work evaluated two implementations of two-box spacer modeling, differing primarily in their calibration approach. One model was calibrated at each width, while the second was calibrated at a nominal spacer width, and resized for each model. The results demonstrate that the accuracy of the simplified spacer geometry depends strongly on the spacer construction and the level at which performance is evaluated. For nominal-width calibrated spacers evaluated at the component level, error increases with spacer width and varies significantly across spacer families. At the window-level, these trends are not uniformly reflected, with some models maintaining a systematic error increase, and others with varying, but not systematically increasing, errors.

For nominal-width calibrated spacers at the component level, absolute U-factor errors increased with spacer width. Higher effective thermal conductivity spacers exhibit both larger error magnitudes and greater sensitivity to spacer width variation. However, the relationship between error and width is not strictly linear. The observed curvature more closely resembles a sub-linear response, suggesting diminishing incremental sensitivity at larger widths. The metal single seal spacer family exhibited a slight decrease in absolute U-factor error at the largest width.

Full-window absolute U-factor error remains near zero (<0.008 $W/m^2 \cdot K$) for the composite single seal and metal single seal spacers across the full width range examined. In contrast, the metal double seal spacer exhibits a steady increase in full-window U-factor error with increasing width, up to 0.03 $W/m^2 \cdot K$. The absolute error growth of the metal double seal spacers follow an approximately linear trend, with $R^2 = 0.98$. This indicates that spacer families with intermediate effective conductivity may be more sensitive to deviations in heat-flow path representation when simplified, whereas low and high conductivity spacers tend to be dominated by a single thermal resistance pathway and are therefore less sensitive to changes in spacer heat-flow paths.

A similar pattern is observed with CI edge results. The composite single seal spacer shows low and slowly varying error across all widths, while the metal double seal spacer exhibits a strong and consistent increase in error magnitude. The metal single seal spacer shows more variability, but without a clear trend. This suggests that CI edge error is particularly sensitive to how well the simplified model captures the balance between spacer conduction and adjacent glazing heat flow, a balance that appears most delicate for spacers with moderate effective conductivity.

CI frame error exhibits the largest absolute deviations among the three metrics and shows the strongest spacer-family dependence. The metal double seal spacer consistently produces the highest CI frame error across nearly all widths, in some cases exceeding an order of magnitude greater error than the composite spacer (composite: 0.10, metal double: 1.56). One plausible explanation is that for very low or very high conductivity spacers, the heat flow is dominated by either the spacer or the surrounding frame/glazing, reducing sensitivity to how the spacer is partitioned in the simplified model. In contrast, spacers with intermediate conductivity occupy a transitional regime in which small changes in modeled heat-flow paths can significantly alter the distribution of heat between the spacer, edge, and frame regions.

The discrepancies in full-window metrics show that extra care and research is needed to fully identify the effects of spacer conductivity on heat flow paths within the window. The results suggest that spacers with an intermediate effective conductivity can induce a state of high instability, where small changes in heat-flow path can drastically change results. At low, and high effective spacer conductivities, these effects are subdued, and heat flow paths are less sensitive to change.

Overall, these results suggest that simplified spacer models can provide accurate full-window performance predictions across a wide range of spacer widths, provided that spacer family characteristics are considered, or that two-box models are calibrated at each spacer width evaluated. Spacer effective thermal conductivity emerges as a key parameter governing both the magnitude and width sensitivity of modeling error. This finding supports the use of family-specific calibration strategies and highlights the importance of validating simplified representations against detailed models across the full range of expected geometric variation.

Limitations

Several limitations should be noted when interpreting these results. Firstly, the simplified spacer models were calibrated at a single nominal width and then applied across a broader width range without further adjustment. While this approach intentionally reflects how simplified models would be used in practice, it necessarily limits accuracy for spacer families whose thermal behaviour varies non-linearly with width. In particular, spacer constructions with intermediate effective thermal conductivity appear more sensitive to changes in spacer geometry, resulting in larger CI edge and CI frame deviations at increased widths.

Second, the analysis focuses on a limited set of spacer families and does not exhaustively represent the full diversity of commercial spacer designs, sealant configurations, or material combinations. While the selected spacers span a meaningful range of effective conductivities, the results should not be interpreted as universally applicable to all spacer systems without additional validation.

Third, error metrics are evaluated relative to detailed numerical models rather than experimental measurements. Although the detailed models represent the current best practice for fenestration thermal analysis, any modeling assumptions inherent in the reference simulations propagate into the reported errors. As a result, the findings quantify consistency with the detailed modeling framework, rather than absolute physical truth.

Finally, the study evaluates spacer behaviour within a specific frame and glazing configuration. Different frame materials, frame geometries, or glazing makeups may alter the relative importance of spacer-edge and spacer-frame interactions, potentially changing the magnitude or distribution of CI errors. Extending the analysis to additional frame and glazing types would further strengthen the generality of the conclusions.

Conclusion

This study evaluated the validity and limitations of two-box spacer modeling across a practical range of spacer widths for insulated glass units. Two modeling strategies were examined: two-box models calibrated independently at each spacer width, and two-box models calibrated at a single nominal width and subsequently resized without recalibration. Both approaches were assessed at the component level and within full-window assemblies for U-factor and CI performance.

Results demonstrate that two-box models calibrated at each spacer width reproduce detailed spacer behaviour with high fidelity across the full evaluated width range. When integrated into complete window assemblies, full-window U-factor, CI edge, and CI frame deviations remained small and stable for all spacer families considered. These findings confirm that width-specific two-box calibration provides a robust and defensible abstraction of detailed spacer geometry for both engineering analysis and regulatory modeling.

In contrast, two-box models calibrated at a single nominal width exhibit increasing sensitivity to spacer width variation, particularly at the component level and for spacer families with moderate to high effective thermal conductivity. While some spacer families maintained acceptable full-window performance across the evaluated range, others showed systematic growth in U-factor and CI error when applied outside the calibration width. This indicates that nominal-width calibration is not generally transferable across spacer widths without loss of accuracy, especially for spacer systems in which lateral heat flow plays a significant role.

Spacer effective thermal conductivity emerged as a primary factor governing both the magnitude of modeling error and its sensitivity to spacer width. Spacer families with intermediate effective conductivity were found to occupy a transitional regime in which simplified representations are most sensitive to geometric variation, leading to amplified CI edge and CI frame deviations. In contrast, low and high conductivity spacers were dominated by a single heat-flow pathway, resulting in reduced sensitivity to spacer abstraction and width variation.

Overall, the results support the use of two-box spacer models as a practical and accurate alternative to detailed spacer geometry, provided that calibration is performed in a manner consistent with the intended application. For analyses requiring robustness across multiple spacer widths, width-specific calibration is recommended. Nominal-width calibration may be suitable for limited applications, but its validity should be verified for each spacer family and performance metric of interest. These findings provide a foundation for more standardized, transparent spacer abstraction methods and support the continued use of simplified spacer models within both North American and European fenestration thermal modeling frameworks. Future work should extend this validation to additional glazing systems, frame systems, and physical hot-box validation tests.

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